

AI-Assisted NOC Incident Analysis & Thematic Monitoring Project

Applying AI-assisted thematic analysis and spreadsheet-based classification to identify recurring infrastructure incidents, operational impact, and proactive NOC monitoring opportunities.

Project Objective

This project demonstrates the application of AI-assisted thematic analysis to a synthetic NOC incident dataset. The objective was to classify infrastructure and operational incidents, identify recurring technical themes, assess operational impact, and derive proactive monitoring recommendations using AI, structured classification rules, and spreadsheet-based analysis.

Dataset

The project uses a synthetic dataset containing **200 NOC incidents** representing common infrastructure and IT operations scenarios, including server availability, network connectivity, application performance, authentication, backup, storage, cloud, and database-related incidents.

The dataset was analyzed using AI-assisted classification followed by structured rule application and human spot-checking.

Methodology

1. Incident Classification

Incident descriptions were analyzed using an LLM to identify the incident topic, category, subcategory, operational impact, customer impact, and suggested cause.

2. Classification Rules

A standardized classification taxonomy was created to ensure consistent categorization across the incident dataset.

3. Human Validation

AI-generated classifications were reviewed against the classification rules, with representative incidents manually spot-checked to identify inconsistencies and validate the approach.

4. Thematic Analysis

Finalized classifications were aggregated to identify recurring incident themes, category distribution, operational impact patterns, and infrastructure areas requiring proactive monitoring.

5. NOC Recommendations

The identified themes were translated into practical monitoring opportunities such as server availability monitoring, CPU utilization monitoring, network interface monitoring, VPN monitoring, backup monitoring, and disk capacity monitoring.

Key Findings

Server incidents were the largest category - 72 of 200 incidents (36.0%) were classified as Server incidents.

Network incidents were the second-largest category - 63 of 200 incidents (31.5%) were classified as Network incidents.

Infrastructure incidents dominated the workload - Server and Network incidents together represented 67.5% of the dataset, indicating that infrastructure and connectivity monitoring are major areas of operational activity.

High-impact incidents were significant - 131 of 200 incidents (65.5%) were classified as High operational impact.

Most common incident theme - Server Unavailability was the most frequent incident theme, with 27 incidents, followed by High CPU Utilization with 23 incidents.

Rank	Incident Theme	Incidents
1	Server Unavailability	27
2	High CPU Utilization	23
3	Backup Job Failure	15
4	Authentication Failure	14
5	Network Interface Down	14
6	VPN Connectivity Failure	14
7	Shared File Service Slowness	11
8	DNS Resolution Failure	11
9	High Disk Utilization	11
10	High Memory Utilization	10
11	High Packet Loss	10
12	Interface CRC Errors	10

Proactive NOC Monitoring Recommendations

Server Availability

Implement proactive health checks and availability monitoring for critical servers to detect failures before they become widespread operational incidents.

CPU Monitoring

Use CPU utilization thresholds and trend analysis to identify sustained resource pressure before performance degradation occurs.

Network Monitoring

Monitor interface status, packet loss, CRC errors, VPN connectivity, and network utilization to identify connectivity degradation.

Backup Monitoring

Monitor backup job status and storage capacity to prevent backup failures caused by insufficient capacity.

Disk Capacity

Implement capacity thresholds and growth monitoring for critical servers and database infrastructure.

Authentication Monitoring

Monitor authentication failures and internal portal availability to identify recurring access issues.

AI Application

AI was used as an analytical assistant rather than as an unrestricted decision-maker. The LLM performed first-pass classification and thematic interpretation of incident descriptions. Structured classification rules were then used to standardize the output, followed by human validation and representative spot-checking.

This approach demonstrates a practical **AI + human-in-the-loop workflow** suitable for operational analysis.

Tools & Technologies

- Cisco Apply AI
- LLM / Generative AI
- Microsoft Excel
- Thematic Analysis
- Data Classification
- Spreadsheet Analysis
- NOC / IT Operations Concepts
- AI Tool Selection & LLM Prompting

Data Limitation

Customer impact was classified as Unknown where the incident description did not provide sufficient evidence to determine customer impact. This prevents unsupported assumptions and demonstrates the importance of distinguishing missing information from confirmed customer impact.

Conclusion

This project demonstrates how AI-assisted analysis can support NOC operations by converting unstructured incident descriptions into structured classifications and actionable operational insights. The analysis identified server and network incidents as the dominant areas of the dataset and highlighted recurring themes such as server unavailability, CPU utilization, backup failures, authentication failures, VPN connectivity, and network interface issues.

The resulting analysis can support proactive monitoring, incident prioritization, capacity planning, and identification of recurring operational problems.